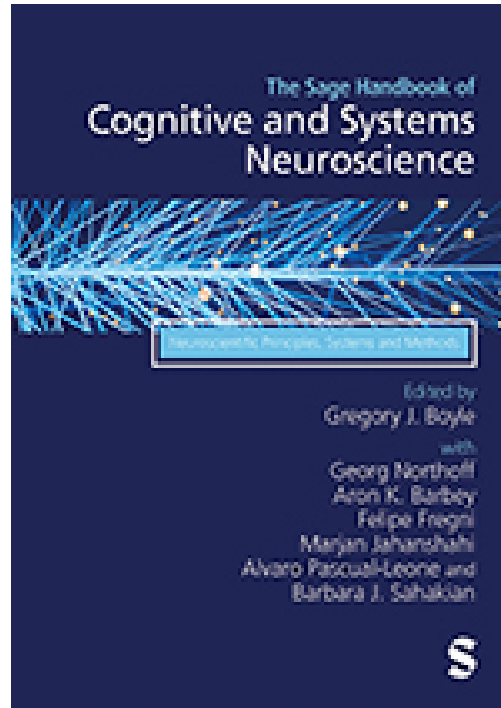


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Cognitive Neuroscience: Basic Principles, Systems, and Methods

Introduction

Cognitive neuroscience is the interdisciplinary study of how cognitive/intellectual functions are processed and represented within the brain (e.g., Braun et al., 2015; Buzsáki, 2019, 2020; Carrion-Castillo et al., 2020; Chen et al., 2019; Costanzo et al., 2015; Donoso et al., 2014; Douw et al., 2016; Gazzaniga, 2018; Gotts et al., 2013; Gross, 1995; McAvoy et al., 2016; Northoff, 2018, 2022; Picchioni et al., 2007; Poeppel et al., 2020; Sahakian, 2014; Sahakian and Gottwald, 2016; Tandon, 2000; Wang et al., 2014; Winters et al., 2004; Wu et al., 2021, 2022a; Zhang et al., 2022). Cognitive neuroscience is the result of merging psychology (i.e., the scientific exploration of human cognitive mental processes), and neuroscience (i.e., the scientific study of the central nervous system or CNS), in order to study how the human brain processes and supports cognitive functions using rather sophisticated technologies (e.g., fMRI; fNIRS; MEG – see below). Since its inception in the early 1990s, cognitive neuroscience has shed substantial light on many such important questions despite the emergence of some critiques (e.g., Uttal, 2011). Essentially, the core belief underlying cognitive neuroscience is that understanding the neurobiological underpinnings of psychological, cognitive, and behavioral processes can offer not only relevant fundamental insights into brain–behavioral relations, but possibly also lead to actionable knowledge that can lead to impactful translational clinical applications and novel therapeutic approaches to patients with various brain-related disabilities.

Historical Traces of Cognitive Neuroscience

Cognitive neuroscience can be traced historically to behavioural neurology and clinical neuropsychology during the nineteenth century wherein patients with brain lesions were investigated (cf. Boyle et al., 2012b, 2023a,b). The study of brain-injured patients has revealed how cognitive functions, such as attention (e.g., visual attention; Kastner and Ungerleider, 2000), perception (Pomerantz, 2003), working memory (Baddeley, 2007; Redondo-Camós et al., 2022; Schacter and Addis, 2007), awareness, and learning (Terry, 2006) are af-

ected by lesions within particular anatomical areas of the brain. Historically, the investigation of brain lesions led French neurologist Paul Broca to discover a distinct brain region in the left inferior frontal cortex that is primarily dedicated to speech production – designated as the speech motor articulation center (Broca's area; cf. Hagoort, 2014). Likewise, Carl Wernicke discovered the predominant sensory speech reception center (Wernicke's area) in the left temporo-parietal cortex, necessary for speech reception and comprehension (cf. DeWitt and Rauschecker, 2013). The application of lesion methods to the study of large-scale brain networks further enabled the characterization of cognitive control networks for adaptive behavior and decision making (Barbey et al., 2012, 2013; Hsu et al., 2022; Ji et al., 2019; Shine et al., 2016), along with neural networks that underlie specific facets of intelligence (cf. Barbey et al., 2014a, 2021; cf. Boyle, 1988; Boyle et al., 2012a; Brody, 1992; Carroll, 1993; Cattell, 1982, 1987; Horn, 1988; Neisser et al., 1996; Sternberg and Kaufman, 1998; Varshney and Barbey, 2021), including the ability factors included in the comprehensive Cattell-Horn-Carroll (CHC) theory of cognitive abilities (Schneider and McGrew, 2018).

In this context of lesion-studies and brain–behavior relations, it is important to recall the seminal contributions of Constantin von Monakov who emphasized the crucial role of the time domain or “chronometry” in understanding the neural substrate on the basis of brain lesions and argued that knowing the location and characteristics of a lesion is necessary but not sufficient in understanding its impact on behavior (von Monakov and Harris, 1969). Instead, von Monakov pioneered the idea that lesions impact activity across distributed networks of the brain (‘functional localization’) and the brain in turn adapts to such functional impact given rise to signs and symptoms (‘symptom localization’). Informed by such notions and leveraging the capabilities of modern brain imaging, resting-state functional magnetic resonance imaging, large data-sets, and sophisticated analytic methodologies, “lesion network mapping” offers a transformative method to relate patterns of brain activity and the functional impact of brain lesions to cognitive processes, behaviors, and symptoms of disease (Boes et al., 2015; Fischer et al., 2016; Joutsa et al., 2022; Kim et al., 2021).

In the development of cognitive neuroscience, the early neuroanatomical discoveries by Broca and Wernicke (cf. Raichle, 2009) and the theoretical elaborations by von Monakov served as paradigmatic examples that have extended into the contemporary connectome mapping of cognitive, affective, and social functions in relation to specific anatomical brain regions/networks (cf. Bassett et al., 2020; Bolognini et al., 2013; Glasser et al., 2013; Neumann et al., 2015, 2016; Toga et al., 2012; Van Essen et al., 2013). Localization of cognitive functions to specific brain regions and networks is distinguished from a more holistic view that assumes global brain activity to be central for cognition (equipotentialism; cf. Halstead, 1947; Hugdahl, 2000). Initially, equipotentiality had been defined by Karl Lashley in the 1950s as the capacity of an intact section of the brain

to carry out functions lost by damage incurred in other areas of the brain (cf. Fancher and Rutherford, 2012).

Another historical origin of cognitive neuroscience was behaviorism at the beginning of the twentieth century wherein psychological functions were explained in terms of their behavior as input–output relations (extended into the modern cognitive science era wherein such functions are accounted for largely in terms of cognitive input–output relations). Behavioral scientists recognized the importance of investigating actual observable behaviors (e.g., Skinner, 1976; Watson, 2017; cf. Carpintero, 2004; Malone, 2014; Rilling, 2000). Behaviorism did not deny the existence of subjective experiences but viewed any conscious or cognitive phenomena (Gage and Baars, 2018; Kosslyn and Andersen, 1995) as depending on the behavior of the organism. As subjective experience (e.g., cognition) was defined as inaccessible for scientific inquiry, this shifted the focus onto behavioristic stimulus–response relationships. Such stimulus–response relationships have been studied with different types of tasks such as simple choice, and go- versus no-go reaction times that have measured and compared response times to a wide range of incoming stimuli (Jahanshahi, 2003). In conjunction with the go- no-go reaction time tasks that allow investigation of action restraint, subsequent development of a stop signal reaction task (Logan and Cowan, 1984) and variants of this have facilitated the study of behavioural inhibition, which is an important component of cognitive control (Jahanshahi et al., 2015).

In recent decades, there has been extensive examination of the brain regions and neural networks involved in the stop signal reaction time task. Aron et al. (2003) discovered that a neural circuit including the right inferior frontal gyrus was responsible for response inhibition. A subsequent study demonstrated a reduction in impulsivity by drugs that enhance both dopamine and noradrenaline activity in the brain, such as methylphenidate, but also drugs which were more selective for noradrenaline, such as atomoxetine (Chamberlain et al., 2006). In a functional magnetic resonance imaging (fMRI) study, Chamberlain et al. (2009) showed that atomoxetine modulated right inferior frontal activation during inhibitory control, thus demonstrating that noradrenaline modulates response inhibition by its actions on the right inferior frontal gyrus.

Current Status of Cognitive Neuroscience

Cognitive psychology merged with neuroscience, leading to the development of cognitive neuroscience as an autonomous discipline in the late 1980s, with many researchers who had been carrying out psychological and behavioral studies now looking for neuroscientific explanations of their findings. During the twentieth century, cognitive psychology emerged as a separate field of study, which together with the development of computational models (Churchland and Sejnowski, 1988, 2017; Dayan and Abbott, 2001) served as a basis

for the subsequent emergence of cognitive neuroscience (the so-called “cognitive revolution” – Boone and Piccinini, 2016; Gazzaniga et al., 2018; Mandler, 2002; Miller, 2003; Poeppel et al., 2020). Now cognition was seen as mediating the behavioral response to sensory input and therefore central to understanding the nature of human behavior (cf. Cuzzolin et al., 2020; Laird et al., 2017; Langley et al., 2022). Other key ideas from the “cognitive revolution” included that while the mind is modular, with distinct neural systems dedicated to specific functions, orchestrated in the service of complex, goal-directed behavior (Pinker, 2002).

Emergence of the “cognitive revolution” opened the door for investigating the multifaceted neural underpinnings of cognitive functions such as attention, working memory, and executive functions, as well as of higher-order processes including conceptualization of the self and consciousness (Chan et al., 2008; Chiao et al., 2009a,b; Chiao and Blizinsky, 2016; Fregni et al., 2005; Gazzaniga, 2018; Northoff, 2014a,b; Northoff et al., 2006; Poeppel et al., 2020; Schaefer and Northoff, 2017), paving the way for the future dialogue with other disciplines such as social cognition and the psychological study of personality (cf. Boyle, 2019; Boyle and Saklofske, 2004a,b; Boyle et al., 2008a,b; 2015; Neumann et al., 2016).

Cambridge Neuropsychological Test Automated Battery

In response to gaining a better understanding of attention, working memory and executive function processes, a computerized battery incorporating a wide variety of objective neuropsychological tests – the Cambridge Neuropsychological Test Automated Battery or CANTAB (camcog.com) – was constructed (Robbins et al., 1994, 1997, 2012). Many CANTAB subtests have since been adapted for neuroimaging (cf. Langley et al., 2023). For example, during the CANTAB paired associates learning task (PAL), at lower levels of difficulty, the hippocampi of MCI patients were activated significantly more than in controls, and significantly less at higher levels of difficulty. Thereby, providing neural validation for the CANTAB PAL test in the early detection of Alzheimer's disease (De Rover et al., 2011). Using the CANTAB intra extra dimensional set shifting task (IED), Langley et al. (2021) showed that the frontal striatal circuits involved in cognitive flexibility were altered in premanifest Huntington's disease. Moreover, development of objective standardized neurocognitive psychometric testing has also aided in the understanding of neuromodulation by neurotransmitters (cf. Pama et al., 2013; Sahakian and Owen, 1992). Mehta et al. (2000) using the CANTAB spatial working memory task (SWM) showed that methylphenidate induced improvements in working memory performance, which was as-

sociated with task-related reductions in regional cerebral blood flow in the dorsolateral prefrontal cortex and posterior parietal cortex. This demonstrated that methylphenidate improved performance through greater efficiency of the neural network underlying working memory. In addition, they found that the beneficial effects of methylphenidate on working memory were greatest in the subjects with lower baseline working memory capacity (e.g., episodic autobiographical memory recall; Roehri et al., 2022).

Role of Emotions and Nonconscious Processes

However, until 1990s, the role of emotions and nonconscious processes remained largely neglected by cognitive neuroscientists. At that time, researchers prioritized the analysis of mental states that could be translated in terms of computational operations, namely perception, learning, and memory, excluding the study of emotions, affects, and nonconscious processes among the research interests of the cognitive neurosciences (Barbey et al., 2014b, 2014c; Damasio, 2000; LeDoux, 2000). Recently, computerized batteries for assessing social and emotional cognition have also been developed. For example, the EMOTICOM battery has not only been used in healthy volunteers (Bland et al., 2016), including with tryptophan depletion (Kanen et al., 2021), but also in patients with depression (Dam et al., 2019) and Huntington's disease (Scahill et al., 2020).

Methodologies and Techniques in Cognitive Neuroscience

In more recent years, leveraging further technological developments, new tools have emerged that capture behavior and cognitive processes with exquisite sensitivity leveraging mobile sensors and multimodal signal acquisition (Gomes-Osman et al., 2021; Libon et al., 2022; Souillard-Mandar et al., 2021). Digital technologies can offer time-efficient brain assessments that uncover issues not otherwise detectable using standard tests. First, they can capture metrics with a much higher sampling rate than a human observer (e.g., eye movements, postural and facial expressions, drawing, writing, and spoken output). Second, they enable the analysis of process by quantifying neurocognitive operations in real time as individuals try to solve a given task (e.g., Boston Process Approach), thus quantifying with great accuracy highly nuanced and subtle behaviors previously unobtainable. Third, data-driven AI-supported analytic approaches can detect subtle signs of brain decline and illness with greater sensitivity and specificity so they can be better managed before they become clinically evident and disabling for the patient. Importantly, such approaches can be deployed in real-life settings, avoiding the need for the stressful and artificial constraints of formal testing conditions.

Cognitive neuroscience has been greatly facilitated by the emergence of cutting-edge brain imaging technologies over recent decades, making it possible to investigate the neural correlates of various cognitive functions including identification of which regions/networks and frequencies in the brain's neural activity are recruited in response to specific cognitive tasks (Bassett and Sporns, 2017). Both single photon emission computed tomography (SPECT), and positron emission tomography (PET) emerged in the 1970s (Portnow et al., 2013). For example, a relatively early study combined the spatial resolution of PET with the temporal resolution of movement-related potentials, to compare the neural substrates of temporal decision-making by comparing self-initiated and externally triggered movements. This comparison revealed that the dorsolateral prefrontal cortex mediated volitional decisions about timing of movements; whereas the activation of the supplementary motor area (SMA) related to preparation (Jahanshahi et al., 1995), and as confirmed in a subsequent PET study of stimulus predictability, the activation of the rostral SMA relates to preparation whereas the caudal SMA is a motor execution area (Jenkins et al., 2000). Subsequently, nuclear magnetic resonance (NMR) became possible – later termed magnetic resonance imaging or MRI (see NMR Progress Report, 1984). Using fMRI, the experimenter can administer a cognitive task that manipulates an independent variable (e.g., the number of items to be maintained in working memory) and measure the Blood-Oxygen-Level-Dependent (BOLD) signal (dependent variable), while simultaneously monitoring brain activity (Ogawa et al., 1992) in specific neuroanatomical regions or networks. While early work using neuroimaging focused on finding “brain lesions” or other biomarkers of pathology, contemporary research is focused on detecting disruptions in the of neurocircuitry connectivity including subtle “brain signatures” associated with certain brain disorders. In addition, high-resolution, high-density electroencephalographic (EEG) methods along with computational analysis of the recorded signals, enables the identification of spatio-temporal patterns of brain activity linked to specific cognitive processes, disease symptoms, and behaviors (Michel and Pascual-Leone, 2020).

Non-Invasive Brain Stimulation (Neuromodulation) Techniques

More recently, non-invasive neuromodulation techniques including transcranial electrical stimulation (tES; Brunoni et al., 2022; Fregni et al., 2005; Woods et al., 2016), and transcranial magnetic stimulation (TMS; Caballero-Villarraso et al., 2022; Jannati et al., 2023; Pascual-Leone et al., 1991, 1999, 2000, 2005, 2011; Redondo-Camós et al., 2022), have expanded our knowledge of brain plasticity (Pascual-Leone et al., 2005) by probing the causal contribution of brain areas to behaviour, moving beyond the correlational methods of fMRI (Boes et al., 2018; Frohlich, 2016) and enabling the dissection of chronometric processes of cognition and the mind (Pascual-Leone et al., 2000; Walsh et al., 2006; Walsh and Pascual-Leone, 2003). Both tES

and TMS can be combined with high spatial and temporal resolution techniques (MEG¹-EEG; EEG-fMRI; EEG-fNIRS²), linking structural and neurochemical data (PET-CT; PET-MRI), or using one modality to alter neurocircuitry, and another to establish *in vivo* the disruptive impact (e.g., fMRI-TMS or tES; TMS-EEG; TMS-fNIRS; EEG-tES; cf. Bolognini and Diana, 2021; Bolognini and Ro, 2010; Farzan et al., 2016; Shafi et al., 2012; Wagner et al., 2007). Such multi-modal association approaches further helped to elucidate human brain networks and the mechanisms by which they process cognition, behavior, and emotion. In an early review of applications of TMS for the study of cognition, the range of questions that could be addressed with TMS included a temporary “lesion” effect to determine whether the contribution of a target area to task performance is essential, identifying the time course of cognitive processes (i.e., time window during which the contribution of a brain area is essential or the relative timing of the contribution of two or more brain areas to task performance), what changes in excitability occur in target brain areas with learning, and investigation of intracortical and transcallosal connectivity (Jahanshahi and Rothwell, 2000). TMS can be reliably and safely applied to disrupt the activity of a given brain region transiently and thus provide insights about the causality of brain activity in behavior. As an early application, Pascual-Leone et al. (1991) showed that repetitive TMS can be used to induce speech arrests and study language networks in health and disease – the Neuronix eXimia System is now cleared by the Food and Drug Administration in the United States for this application. Similar application of “temporary lesion approaches” using the controlled perturbation induced by TMS and guided by brain imaging, has served to clarify, for example, the causal role of primary visual cortex (Kosslyn et al., 1999), and the dynamics of interhemispheric compensatory processes (Sack et al., 2005), during mental imagery; or the need for fast back-projections in visual awareness (Pascual-Leone and Walsh, 2001).

Subsequent electrophysiological research has employed TMS not only for establishing functional specialization, but also for examining functional connectivity, afunctional interaction, and integration within the cognitive domain (Jahanshahi, 2005). Other applications of TMS to the investigation of cognition have included the pharmacology of cognition through the study of the effects of medication, the impact of surgical or rehabilitative interventions, as well as brain damage, and/or neurological or psychiatric disorders on cognition (Jahanshahi, 2005; Jahanshahi and Rothwell, 2000). Subsequent applications of TMS as a therapeutic tool have included increasing processing speed, improving selective attention or naming, improving accessibility to memory, or enhancing implicit learning (Bolognini and Miniussi, 2018; Jahanshahi, 2005; Vallar and Bolognini, 2011). In addition, in treatment resistant patients with obsessive-compulsive disorder, deep brain stimulation (DBS) has been shown to improve cognitive flexibility (Tyagi et al., 2019). The past 25 years has

brought substantial advances in knowledge about the mechanisms of action of non-invasive brain stimulation; novel and more precise technologies; safety, training, regulatory and ethical guidelines along with research applications and their clinical translation. New non-invasive brain stimulation methods and techniques, such as Temporal Interference Stimulation (TIS) enable more precise and even selective non-invasive stimulation of deep brain structures (Grossman et al., 2017).

Decade of The Brain: Computational Approaches and Cognitive Neuroscience Studies

The decade of the brain (Jones and Mendell, 1999) has inspired extensive brain mapping efforts worldwide, including the Human Brain Project in Europe (HBP; Markram et al., 2011; Sahakian et al., 2021), the ENIGMA Consortium (Kong et al., 2018), the Human Connectome Project (HCP; Toga et al., 2012; cf. Glasser et al., 2013; Van Essen et al., 2013), the China Brain Project (CBP; Poo et al., 2016), and the BRAIN Initiative (BI; Litvina et al., 2019). Some of the highlights of the HBP include the development of the Julich-Brain Atlas (JBA; Amunts et al., 2020) and the Virtual Epileptic Patient (VEP; Jirsa et al., 2017). The European-funded HBP has also created EBRAINS, an open research infrastructure built on the scientific advances and tools developed by the project's research teams, and making them available to the scientific community via a shared digital platform (<https://ebrains.eu/>).

Computational tools have been developed and Big Data has been compiled (Fox and Lancaster, 2002; Yarkoni et al., 2011), as well as the recent standardization Neuroimaging Meta-Analysis Research Environment (NiMARE)³ endeavor. The aim has been (1) to link functional and structural brain maps to cognitive tasks from previous studies and (2) to amass and share large data sets. These large-scale collaborations have enabled application of artificial intelligence techniques using machine learning to predict behavior from measures of brain activity (Li et al., 2019). The combination of Big Data from neuroimaging with the human genome project and genetic neuroimaging provide remarkable opportunities for the next decade of research and discovery in cognitive neuroscience. Even more powerful is when these Big Data cohorts include neuroimaging, cognitive and behavioural measures, such as in the IMAGEN, ABCD, and UK Biobank research projects. Recent publications from these large cohorts have investigated many aspects of brain and behav-

ioral function, including sleep (Li et al., 2022), social isolation (Shen et al., 2022), attention deficit hyperactivity disorder (Shen et al., 2020a,b), OCD (Wu et al., 2022b), and the androgynous brain (Zhang et al., 2021).

Comparative Cognitive Neuroscience Including Studies Into Infra-Human Species

The study of non-human primates is also an area of interest in cognitive neuroscience that examines our similarities with respect to cognitive capabilities, physiology, neuroanatomy, social complexity, reproduction, and development. Recent developments in magnetic resonance imaging, for instance, have the potential for extracting neurobiological measures that can be compared between species within a phylogenetic framework. Comparative cognitive neuroscience enriches the understanding of how the human brain works and the appreciation of its uniqueness by looking to problems the brain evolved to solve and how these relate to its organization (Mars et al., 2014). Both historical lines of development involving the neuroscientific study of the human brain and of the brains of non-human primates allows comparative cognitive neuroscience to share certain empirical features and philosophical assumptions.

Spatiotemporal Cognitive Neuroscience

However, both the human and non-human primate models of cognitive neuroscience have certain limitations with respect to mental features like self and consciousness (e.g., Chiao et al., 2009a,b) making necessary its extension into what has been termed “spatiotemporal cognitive neuroscience.” Most cognitive neuroscientists agree that we need to link mental operations to brain functions and to do so within an interdisciplinary framework that appreciates the importance of both experimental cognitive psychology and neuroscience. We emphasize the need for approaches that seek to integrate experimental psychology with cognitive neuroscience, such as spatiotemporal cognitive neuroscience (Northoff and Hirjak, 2022; Northoff and Huang, 2017; Northoff et al., 2020a,b), as well as cognitive computational neuroscience (Kriegeskorte and Douglas, 2018).

Summary and Conclusions

Over recent years, cognitive neuroscience has been greatly facilitated by development of cutting-edge brain imaging technologies, enabling investigation of the neural correlates of various cognitive functions and the identification of which regions/networks in the brain's neural activity are recruited in response to specific cog-

nitive tasks (Bassett and Sporns, 2017). The decade of the brain (Jones and Mendell, 1999) has inspired extensive brain mapping efforts worldwide, including the Human Brain Project in Europe (HBP; Sahakian et al., 2021), the ENIGMA Consortium (Kong et al., 2018), the Human Connectome Project (HCP; Toga et al., 2012), the China Brain Project (CBP; Chang et al., 2021), and the BRAIN Initiative (BI; Litvina et al., 2019). Some of the highlights of the HBP include the development of the Julich-Brain Atlas (JBA; Amunts et al., 2020) and the Virtual Epileptic Patient (VEP; Jirsa et al., 2017). It has also created EBRAINS, an open research infrastructure available to the scientific community via a shared digital platform (<https://ebrains.eu/>). Computational tools have been developed and Big Data has been compiled (Yarkoni et al., 2011), as well as the recent Neuroimaging Meta-Analysis Research Environment (NiMARE) endeavor.

Combining Big Data from neuroimaging with the human genome project and genetic neuroimaging (e.g., Wu et al., 2022b) has enabled remarkable opportunities for future research and discovery within cognitive neuroscience. These Big Data cohorts include neuroimaging, cognitive and behavioural measures, such as in the IMAGEN, ABCD, and UK Biobank research projects. Recent publications from these large cohorts have investigated many aspects of brain and behavioral function, including sleep (Li et al., 2022), social isolation (Shen et al., 2022), attention-deficit hyperactivity disorder (Shen et al., 2020a,b), obsessive-compulsive disorder (Wu et al., 2022b), and the androgynous brain (Zhang et al., 2021).

One point we wish to emphasize is that even though the cognitivist paradigm reintroduced the notion of “mind” into experimental cognitive psychology, it led cognitive neuroscience into mutually exclusive paradigms, both philosophically and empirically, which have resurfaced in terms of the distinction between the brain's neuronal activity versus cognition. Recent developments have expanded the spatiotemporal framework of cognitive neuroscience by intrinsically integrating both neuronal activity and cognition, into the current approach of “Spatiotemporal Neuroscience.” Clearly though, we consider that there is need for approaches that seek to integrate, for example, cognitive neuroscience with spatiotemporal neuroscience, along with cognitive computational neuroscience.

Notes

¹ MEG: Magnetoencephalography (a non-invasive functional neuroimaging technique) measures magnetic fields at the scalp resulting from electrical brain activity.

[2](#) fNIRS: functional Near-Infrared Spectroscopy (a non-invasive optical imaging technique) measures brain changes in hemoglobin concentrations upon light being shone through the skull and refracted back to a specialized detector (optode).

[3](#) NiMARE provides workflows as command-line interfaces, including ALE meta-analysis, meta-analytic coactivation modeling (MACM) analysis, peaks2maps image reconstruction, and contrast map meta-analysis. Each workflow generates a boilerplate paragraph with workflow details and relevant citations.

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